

Contact processes on ballistic Poisson particles: criticality, fronts, and when motion helps colonization

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Abstract

和文要旨. 実測恒星運動学を動機として、等速直線運動する Poisson 粒子系上の接触過程 $M(\rho, \nu, R, p, \tau, T_s)$ (進入駆動・撃ちっ放し・対一回・再マーク許可) を導入し、その臨界構造を確立する。主結果: (i) 転送核のスペクトル条件 $r(K) < 1$ による絶滅定理 (定理 A)、(ii) この SIR 型分枝上界は再マーク許可の完全モデルには拡張されないこと (逆向きチャンネルによる期待総子孫数 — 種を含むマーク事象総数 N — の分枝上界超過、 4.4σ の数値確定)、(iii) 静的極限とボンド希釈ランダム幾何グラフの生存事象の同値 (定理 B)、(iv) 完全モデルの生存定理 (定理 C2 — セル・リレー構成、私有コイン結合、正規化橋、区間演算による機械認証)、(v) 「静的幾何が禁じる疎密度でも十分速い運動は入植を可能にする」 (定理 D)、(vi) 前線の超線形/線形二分 (定理 C6a/C6b')。さらに伝播規約への依存性 (進入駆動では運動が単調に助け、滞在時間要求型では最適速度が存在する) を数値的に確立する。検証は独立実装の数値反証装置・新規 AI インスタンスによる 11 サイクルの敵対的審査・幾何定数の機械認証の三層で行い、その全記録を開示する (付録 B)。

Abstract. Motivated by observed stellar kinematics, we introduce a contact process on particles moving ballistically in $\mathbb{R}^3$: a Poisson point process of intensity ρ whose points carry i.i.d. velocities drawn from ν , with entry-driven, fire-and-forget transmission at range R (success probability p , delay τ , each ordered pair attempts at most once ever), lifetimes $\text{Exp}(T_s)$, and re-marking of dead points allowed. We establish the critical structure of this model. First, an extinction theorem: if the spectral radius of an explicit velocity transfer kernel is smaller than one, the SIR-type variant dies out almost surely (Theorem A), and the underlying moment computation is exact. Second, this branching-type upper bound *does not extend* to the full model with re-marking: a “reverse channel”, by which a child re-marks its dead parent, produces a first-order excess of the expected total progeny (mark events including the seed) over the branching bound, certified numerically at 4.4σ (Section 4). Third, in the static limit the survival event coincides with percolation of a bond-diluted random geometric graph (Theorem B). Fourth, our main result: a survival theorem for the full model (Theorem C2), proved by a cell-relay renormalization whose geometric constants are certified by interval arithmetic, together with a private-coin coupling lemma, a disclosure-capsule identification, and a normalization bridge between accepted and disclosed candidate sets. Fifth, combining Theorems B and C2: at densities where static geometry forbids percolation for every p , sufficiently fast motion enables colonization (Theorem D). Finally we prove a superlinear/linear front dichotomy (Theorems C6a, C6b'), and establish numerically a *convention dichotomy*: under entry-driven transmission motion helps monotonically, while under dwell-time requirements an optimal speed emerges.

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All proofs were developed under a three-layer verification protocol (independent numerical falsification device, eleven cycles of adversarial line-by-line review by fresh AI instances, machine certification of all proof geometry), fully disclosed in Appendix B.

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1 Introduction and main results

Consider a population of particles scattered in $\mathbb{R}^3$ as a Poisson point process, each moving forever along a straight line with a random velocity. A single marked particle starts an epidemic: whenever an unmarked particle *enters* the ball of radius R around a living marked particle, the mark is transmitted with probability p after a travel delay τ ; marks die at rate $1/T_s$; dead particles can be re-marked. Does the epidemic survive forever? How fast does it spread? And — the question that motivated this work — *does the motion of the particles help or hinder?*

The model is a mathematical idealization of interstellar settlement fronts carried by probes among moving stars, in the convention of Carroll-Nellenback et al. [3] (“entry-driven, fire-and-forget”: a settled system launches a probe when an unsettled system comes within range, and the probe travels once). The application of the resulting theory to observed stellar catalogs is deliberately separated into companion work; the present paper is self-contained probability theory.

Our model sits in a genuine gap between two literatures: epidemics on *diffusing* populations (Brownian motions with removal, Grimmett–Li [6]; spread of infection among random walks, Kesten–Sidoravicius [8, 9], Dauvergne–Sly [4]) and *static* continuum percolation (random geometric graphs, Penrose [16]; Poisson cylinders, Broman–Tykesson [2]). Ballistic (constant-velocity) motion is the kinematically correct regime for stars on $\pm$ few-Myr horizons, and it changes the mathematics: the space-time track of a particle is a straight cylinder, contacts between any ordered pair form a *single* time interval, and the environment de-correlates by transport rather than by mixing. A space-time percolation picture of this kind appears, non-rigorously, in delay-tolerant-network engineering [7]. Monotonicity of criticality in the velocity scale is open even in neighbouring models (jump-rate monotonicity for spread of infection; environment-speed monotonicity in evolving-environment contact processes, Linker–Remenik [14]), and is known to be direction-dependent across settings. Infections with recovery among moving particles are treated in [1]; percolated (bond-diluted) random geometric graphs, the static skeleton of our model, in [5, 11].

The model $M(\rho, \nu, R, p, \tau, T_s)$

At time 0, particles form a homogeneous Poisson process of intensity ρ on $\mathbb{R}^3$; particle i receives an i.i.d. velocity $v_i \sim \nu$ and moves as $x_i(t) = x_i + v_i t$ (the configuration is time-stationary). For an ordered pair (i, j) the contact set $\{t : |x_i(t) - x_j(t)| \leq R\}$ is a single interval $[\alpha_{ij}, \beta_{ij}]$. Transmission is *entry-driven and fire-and-forget*: if i is marked and alive during its contact window with j , then at time $a_{ij} = \max(\alpha_{ij}, m_i)$ (provided $a_{ij} < d_i$ and $a_{ij} \leq \beta_{ij}$) the pair attempts transmission *once and only once ever* (pair-once is an absolute convention; re-marking never re-opens a pair), succeeding with probability p ; on success j becomes marked at time $a_{ij} + \tau$. Mark lifetimes are $\text{Exp}(T_s)$; a particle whose mark has died may be marked again by a fresh pair (*re-marking*). The seed is one marked particle near the origin at $t = 0$. *Survival* means the set of living marked particles is non-empty for all times.

Two dimensionless quantities organize the phase diagram: with $\lambda := \rho \pi R^2 \mathbb{E}|v - v'|$ (entry

rate; $v, v' \sim \nu$ independent) and $\deg := \frac{4\pi}{3}\rho R^3$,

$$m := p(\deg + \lambda T_s), \quad K_{\text{mix}} := \frac{w_1 T_s}{R} \quad (\text{mixing number of a speed shell } w_1).$$

Main results

Write $r(K)$ for the spectral radius on $L^2(\nu)$ of the transfer kernel

$$K(v, v') = p\rho \left(\frac{4\pi}{3}R^3 + \pi R^2 |v - v'| T_s \right).$$

Theorem A (extinction, SIR variant). *Assume $\mathbb{E}|v|^2 < \infty$. If $r(K) < 1$, the no-re-marking (SIR) variant dies out almost surely. The underlying first-moment (Mecke) computation of the expected number of length- k transmission chains is exact, so the spectral condition is sharp at the level of first moments.*

Proposition (the SIR bound does not extend; Section 4). *There are parameter regions with scalar mean $\bar{m} < 1$ in which the full model satisfies $\mathbb{E}[N] > 1/(1 - \bar{m})$, where N is the total progeny: the total number of mark events, including the seed. Consequently, the SIR-type (branching-process) threshold description does not extend to the full model with re-marking: the excess is produced by a first-order “reverse channel” (a child re-marks its dead parent through the unused reverse pair). Whether the extinction threshold itself separates — i.e., whether the full model can survive somewhere inside $r(K) < 1$ — is open (Section 10).*

Theorem B (static limit). *Let $\nu = \delta_0$. Then the survival event of $M(\rho, \delta_0, R, p, \tau, T_s)$ coincides (up to null sets) with the event that the origin’s cluster is infinite in the bond-diluted random geometric graph $G(\rho, R; p)$, for every T_s and τ . In particular survival does not depend on T_s, τ in the static limit.*

Theorem C2 (survival of the full model). *Let ν be isotropic with mass $q > 0$ on a speed shell $S = \{w : |w| \in [w_1, 2w_1]\}$, and with a density on S bounded by $\bar{\Lambda} \cdot q/\text{vol}(S)$ with $\bar{\Lambda} \leq 4$ (relative density ratio; Maxwellian ν satisfies $\bar{\Lambda} \approx 2.4$ on its typical shell). Normalize $R = 1$ and assume the mixing condition $K_{\text{mix}} = w_1 T_s \geq K_0 = 20$. There is a constant $C_0 = C_0(\tau/T_s) < \infty$ such that*

$$m_1 := p\rho q \pi w_1 T_s > C_0 \quad \implies \quad \text{the full model survives with positive probability.}$$

For general $\bar{\Lambda} < \infty$ the same holds with $K_0 = K_0(\bar{\Lambda}) < \infty$ and $C_0 = C_0(\tau/T_s, K_0(\bar{\Lambda}))$.

Theorem D (motion beats static geometry). *Fix any $0 < \rho < \rho_c^{\text{RGG}}(p=1)$, so that by Theorem B the static model dies out almost surely for every $p \leq 1$, T_s , τ . Then for every $p > 0$, $\tau \geq 0$, $T_s > 0$ there are isotropic velocity laws ν (scaled copies ν_s of any law satisfying the hypotheses of Theorem C2; the shell mass q and the ratio $\bar{\Lambda}$ are scale-invariant) such that the full model survives with positive probability. Under the entry-driven convention, sufficiently fast motion enables colonization at densities where static geometry forbids it.*

Theorem C6a (superlinear front). *If $T_s = \infty$ and ν has unbounded support, then the front radius satisfies $R(t)/t \rightarrow \infty$ almost surely.*

Theorem C6b' (linear bound for bounded speeds). *If $|v| \leq w_{\max}$ a.s. and $\tau > 0$, then for all $t \geq 0$, $R(t) \leq w_{\max}t + R\lfloor t/\tau \rfloor + |x_0|$, where x_0 is the seed position. In particular the front is at most linear, and $\tau > 0$ is precisely what guarantees well-foundedness of ancestries (Section 8).*

Numerically we complement the dichotomy: unbounded (Gaussian) velocity laws produce an accelerating front (fitted growth exponent 1.34 ± 0.14) while bounded laws are linear (0.98 ± 0.01); and we establish the *convention dichotomy*: under the entry-driven convention survival probability is monotone increasing in the velocity scale, while under a dwell-time-requirement convention an optimal speed $\sigma^* \approx 0.4$ emerges (Section 8).

Method of verification

Every claim in this paper was developed under a claim-driven verification protocol: an independent simulation device for numerical falsification, adversarial line-by-line review of each completed proof by fresh AI instances with no shared context (eleven review cycles for Theorem C2), and machine certification, by interval arithmetic, of every geometric constant in the survival proof. Section 9 states the exact boundary of what this internal verification does and does not establish; Appendix B is the complete method record, including the error ledger.

2 Model and conventions

We collect the conventions used throughout; they match the simulation device of Appendix A exactly.

Dynamics. Poisson(ρ) initial positions on $\mathbb{R}^3$; i.i.d. velocities $v_i \sim \nu$; ballistic motion $x_i(t) = x_i + v_i t$. At every time t the positions again form Poisson(ρ) (time stationarity). Since relative velocities are constant, the contact set of an ordered pair is a single interval; the entry time $a_{ij} = \max(\alpha_{ij}, m_i)$ lies in the contact window whenever an attempt occurs, so the transmission jump in space is at most R .

Transmission conventions. *Entry-driven*: attempts are triggered by entry events (or by the marking of i if j is already inside range). *Fire-and-forget*: no dwell requirement — the attempt is resolved instantly by a fair p -coin, the effect arriving after delay τ . *Pair-once*: each ordered pair (i, j) attempts at most once in the entire history; this is an absolute convention, unaffected by re-marking. *Re-marking*: a dead particle is eligible for fresh marks through fresh pairs. Lifetimes are $\text{Exp}(T_s)$, independent of everything else; marks never kill and never displace — the process is purely additive on marks.

Scaling. $R = 1$ may be assumed (space-time scaling); m , m_1 , K_{mix} , τ/T_s , and the ratio $\bar{\Lambda}$ are scale-invariant. In particular the pushforward $\nu_s = \nu^0(\cdot/s)$ of a shell law keeps q and $\bar{\Lambda}$ fixed while scaling K_{mix} and m_1 linearly in s — the scale-invariance lemma behind Theorem D.

3 Extinction: Theorem A

The proof (Appendix C.1) is a first-moment argument made exact. For a time-ordered chain of transmissions $0 \rightarrow i_1 \rightarrow \dots \rightarrow i_k$, conditioning on velocities and applying the multivariate Mecke equation [10] for the Poisson process of (position, velocity) marks reduces the expected number of such chains to a k -fold operator power. Two technical points make the computation exact rather than merely an upper bound.

First, the *capsule volume lemma*: for a source of velocity v and window length T , the set of initial conditions (x_0, v') whose trajectory enters the moving ball within the window is a capsule of exact volume $\frac{4\pi}{3}R^3 + \pi R^2|v - v'|T$; no overcounting occurs because entry times are unique (single contact interval).

Second, the *deterministic attempt-time recursion*: along a chain, the j -th attempt time satisfies $t_j = \max(\inf T_j, t_{j-1} + \tau)$, a deterministic function of the geometry, which quantifies the windows without conditioning on the past — this removes the window-quantification gap found in review (Appendix B, cycle 1).

Chains do *not* factor into scalar means: the velocity of the relay particle is shared between consecutive links, so the correct object is the kernel K acting on $L^2(\nu)$, not the scalar $\bar{m} = \int K d\nu d\nu$. If $r(K) < 1$, the expected total progeny is finite and the SIR variant dies out a.s.

4 The SIR-type bound does not extend to re-marking

It is tempting to expect Theorem A to hold verbatim for the full model with $\bar{m}$ -type conditions. It does not.

Proposition (reverse channel). *There exist parameters with $\bar{m} < 1$ (e.g. near-static pairs, $p = 1$, $\tau = 10T_s$, $\deg = 0.1$) for which the full model satisfies $\mathbb{E}[N] > 1/(1 - \bar{m})$, where N is the total progeny (mark events including the seed).*

The mechanism is a *first-order self-influence* that Mecke chain counting cannot see: a child j , marked via (i, j) , can re-mark its *dead* parent i through the never-used reverse pair (j, i) — the probability per infection is of order $p(1 - e^{-2\tau/T_s})$ and the channel lives in the corner of slow relative velocities and $\tau \gtrsim T_s$. The device certifies the violation: over 10^4 independent realizations in the counterexample regime, $\mathbb{E}[N] = 1.1353 \pm 0.0055$ against the branching bound $1/(1 - \bar{m}) = 1.1111$, a 4.4σ excess (Figure 1). We emphasize the exact strength of the statement: this is a certified violation of the *expectation bound*, hence of the SIR-type threshold *description*; it does not by itself prove survival anywhere inside $r(K) < 1$, and the separation of the extinction thresholds themselves remains open (Section 10).

5 Static limit: Theorem B

Let $\nu = \delta_0$. Then all contact windows are $[0, \infty)$ or empty, every attempt happens at the instant its source is marked, and delays only shift the marking times. The set of particles ever marked is exactly the cluster of the seed in the graph whose vertices are the Poisson points and whose edges are pairs at distance $\leq R$ retained independently with probability p — the bond-diluted random geometric graph $G(\rho, R; p)$ [16, 15, 5]. The *survival event* therefore coincides a.s. with the percolation event of the seed’s cluster, for every T_s, τ ; lifetimes only govern how

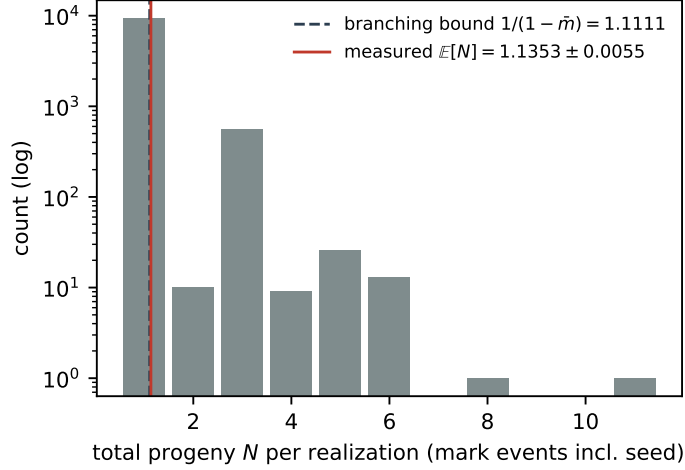


Figure 1: Reverse channel (experiment E8c): histogram of the total progeny N (mark events including the seed) over 10^4 realizations in the counterexample regime. The measured mean exceeds the branching bound by 4.4σ . Reproduce: `python3 src/wake_r/paper_figs.py` (data: `data/phase_r/E8c_counterexample.json`).

long the (predetermined) cluster stays lit. A double numerical verification (dynamic device vs. independent union-find implementation) is recorded in Appendix A.

6 Survival: Theorem C2

This is the technical core of the paper. Full proofs are in Appendix C.3; here we present the architecture and the five load-bearing lemmas. Constants below refer to the certified configuration $K_0 = 20$, $T_a = T_s/32$, $T_g = 64(\tau + T_s)$.

6.1 Cell-relay construction

By the pigeonhole principle, one of the 200 equal-width sub-shells of $[w_1, 2w_1]$, written I^* , carries mass $\geq q/200$; let $\bar{w}$ be its midpoint. The *aiming constant* $\bar{u}_x := \bar{w}/\sqrt{17/16}$ makes the ideal required velocity $(\bar{u}_x, \pm\bar{u}_x/4, 0)$ have norm exactly $\bar{w}$. Sites (n, k) , $n + k$ even, of a two-dimensional oriented lattice are realized as spatial boxes of side $2a$, $a = \bar{u}_x T_g/32$, with y -tracks spaced $8a$ apart and a single z -track; anchoring is *hybrid* (y, z global, x relative to the parent, targets rounded to the lattice $(a/8)\mathbb{Z}^3$). A site is *open* if the relay delivered at least N_h marked, living particles with velocities in a small cover box into the box core (half-width $a/2$) at time t_k , at least one of which has residual lifetime $\geq T_g$ (*coverage clause* — this keeps the marked set non-empty between epochs). From a good parent P^* , the relay uses entries into its radius-1 capsule during the *entry window* $[t_k, t_k + T_a]$ only.

6.2 The five lemmas

1. **Geometry (machine-certified).** Over all perturbations (parent position in core + rounding, parent velocity in the $\pm 0.05\bar{w}$ cover, entry offset $\leq R$, window sweep $s \in$

$[0, T_a]$, branch signs), the landing window $w_c \oplus [-h_0, h_0]^3$, $h_0 = \frac{a/2}{T_g - s}$, retains a margin $h_0 - \text{dev} - I_h - \text{lip} \geq s^* = 0.0048 \bar{w}$ (at $\tau = 0$; larger for $\tau > 0$), stays inside the velocity cone and inside the parent cover. Certification is by interval arithmetic with exact norm bounds and a Lipschitz penalty on an s -grid (Appendix A).

2. **Erosion patch.** With $\theta^* = s^*/(1.05\bar{w})$, the full angular patch of radius θ^* around the aim direction, times the whole radial band I^* , lands in the window: by isotropy, $\hat{\nu}(\text{accepted}) \geq \frac{1}{200} \cdot \frac{1 - \cos \theta^*}{2}$.
3. **Disclosure capsule identification.** The construction reads the randomness of *window entrants only*: the single-contact-interval property makes a window entry a *first* contact, so pair-once guarantees the attempt fires inside the window; entries outside windows never enter the construction's σ -algebra. Hence the deduction tubes below have segment length exactly $|v - v_P|T_a$ — disclosure geometry, not dynamical exposure.
4. **Normalization bridge (machine-certified).** Candidates contaminated by past sites are those in the sets $\{v : \bar{\Delta} - v j T_g \in \mathcal{K}\}$, $\mathcal{K} = B(0, 2R) \oplus \text{seg} \oplus \text{seg}$; the union over the window is contained in a *box-hull* $B(0, 2R) \oplus H_1 \oplus (-H_2)$ whose exact Steiner volume is $\prod c_i + 2r \sum_{i < j} c_i c_j + \pi r^2 \sum c_i + \frac{4\pi}{3} r^3$ ($r = 2R$). With the site-count bound $n_j \leq \frac{\sqrt{3}}{8} j + 1 + \text{diam } \mathcal{K}/(8a)$ (single z -track; gap j fixes the generation) and the density hypothesis, the disclosed mass satisfies $\bar{\Lambda} \hat{\nu}(\text{bad}) \leq \frac{1}{2} \hat{\nu}(\text{accepted})$ for $K_0 = 20$, $T_a = T_s/32$, $\bar{\Lambda} \leq 4$: at $\tau = 0$ (worst case) the certified values are $\hat{\nu}(\text{bad})|_{\bar{\Lambda}=1} = 1.69 \times 10^{-9}$ versus $\hat{\nu}(\text{accepted}) = 2.64 \times 10^{-8}$, i.e. the bridge closes up to $\bar{\Lambda}_{\max} = 7.8$, about twice the assumed bound.
5. **Private-coin coupling (Lemma X).** Conditional on the entire disclosure, the failure of a pair-fresh candidate to be marked-and-alive at the epoch end requires the shortfall of a *single private exponential coin* of that candidate; the conditioning reads lifetimes only through indicator paths $(\mathbf{1}\{L > s\})_{s \leq a}$, under which the relevant residuals remain jointly independent $\text{Exp}(T_s)$; and post-arrival interference can only *rescue* (re-marking never harms — additivity). Hence the number of delivered children stochastically dominates a Binomial with success probability $\beta = e^{-T_g/T_s}$ over a Poisson number of candidates with mean $\mu \geq c_4(\tau/T_s) m_1/32$, surviving the bridge deduction at least in half.

6.3 Assembly

Chernoff bounds give $\mathbb{P}(\text{site open} \mid \text{all disclosure}) \geq 1 - \varepsilon(m_1)$ with $\varepsilon \rightarrow 0$ as $m_1 \rightarrow \infty$; by the tower property this holds sequentially in any breadth-first order, so the open sites dominate an i.i.d. Bernoulli($1 - \varepsilon$) field on the oriented lattice (direct sequential conditioning; cf. the domination theory of [13], which our argument does not need). The comparison lattice is precisely $\mathcal{L} = \{(n, k) \in \mathbb{Z} \times \mathbb{Z}_{\geq 0} : n + k \text{ even}\}$ with adjacency $(n, k) \rightarrow (n \pm 1, k + 1)$ — *the same adjacency structure* for which Liggett's bound $p_c^{\text{site}} \leq 3/4$ is proved [12]; we verified the identification of the two lattices explicitly (each site has exactly the two children $(n \pm 1, k + 1)$ and the two parents $(n \mp 1, k - 1)$, and openness is a site property). For $m_1 > C_0$ we have $1 - \varepsilon > 3/4$, hence supercritical oriented site percolation, an infinite open cluster with positive probability, and — through the coverage clause — a marked living particle at every time. Ignition from the arbitrary-velocity seed is a positive-probability bootstrap. $\square$

Remark 1. No upper bound on m_1 is used anywhere: the coverage clause, the Chernoff step and the observation truncation $M = \lceil 2\mu \rceil$ all improve or are unaffected as m_1 grows. Intermediate

τ are covered by monotonicity of both sides of the bridge. The pure-isotropy case (no density bound) is left as a conjecture (Section 10).

7 Motion beats static geometry: Theorem D

Fix $0 < \rho < \rho_c^{\text{RGG}}(p=1)$. By Theorem B and the standard monotone coupling in p , the static model dies for all $p \leq 1, T_s, \tau$. Now fix any $p > 0, \tau \geq 0, T_s > 0$ and any base law ν^0 satisfying the hypotheses of Theorem C2. Scaling $w \mapsto sw$ preserves the shell mass q and the relative density ratio $\bar{\Lambda}$ (the density scales by s^{-3} and the shell volume by s^3), while $K_{\text{mix}} = s w_1^0 T_s$ and $m_1 = p \rho q \pi s w_1^0 T_s$ grow linearly in s with τ/T_s fixed; for s large both hypotheses of Theorem C2 hold, and the full model survives with positive probability. $\square$

8 Fronts and the convention dichotomy

8.1 Superlinear and linear fronts

Theorem C6a (Appendix C.4) shows $R(t)/t \rightarrow \infty$ a.s. for $T_s = \infty$ and unbounded ν : the seed alone recruits carriers of ever-larger speeds, an infinite-system version of the fastest-of- N heuristic used in [3]. Conversely, Theorem C6b': for $|v| \leq w_{\text{max}}$ and $\tau > 0$,

$$R(t) \leq w_{\text{max}} t + R\left\lfloor \frac{t}{\tau} \right\rfloor + |x_0|.$$

The proof is a strict accounting: along any ancestry of mark events, marking times increase by at least τ per hop — which simultaneously proves the ancestry is *well-founded* (no infinite descent; this, not locality per se, is the true role of $\tau > 0$) — and travel time telescopes exactly across $[0, t]$, each hop contributing a spatial jump $\leq R$ at its attempt time. The companion pathology: at $\tau = 0$, if the open cluster of the seed in the p -bond-diluted instantaneous geometric graph is infinite, the whole cluster is marked at time 0^+ and $R(0^+) = \infty$ can occur.

Numerically the dichotomy is sharp (Figure 2): Gaussian velocity laws accelerate (fitted exponent 1.34 ± 0.14), bounded laws are linear (0.98 ± 0.01).

8.2 The convention dichotomy

Does motion help colonization? Theorem D answers “yes, decisively” for the entry-driven convention, and experiment E4c confirms strict monotonicity of survival probability in the velocity scale ($0.73 \rightarrow 0.91$ over $\sigma \in [0, 0.16]$ at fixed ρ, p). But replace fire-and-forget with a *dwelt-time requirement* (transmission requires residence within range for a fixed time) and the answer inverts: experiment E7 exhibits a non-monotone survival curve with an optimal speed $\sigma^* \approx 0.4$ (survival fraction $0.06 \rightarrow 0.31 \rightarrow 0.00$), consistent with the optimal-velocity phenomenology observed in mobile-agent epidemic models [17, 18]. *The answer to “does motion help” is a function of the transmission convention*, exhibited here within a single model family (Figure 3). A related trio of non-monotonicities (in particle number, in re-marking, in timing — all SIR-type interference effects) is recorded in the method archive.

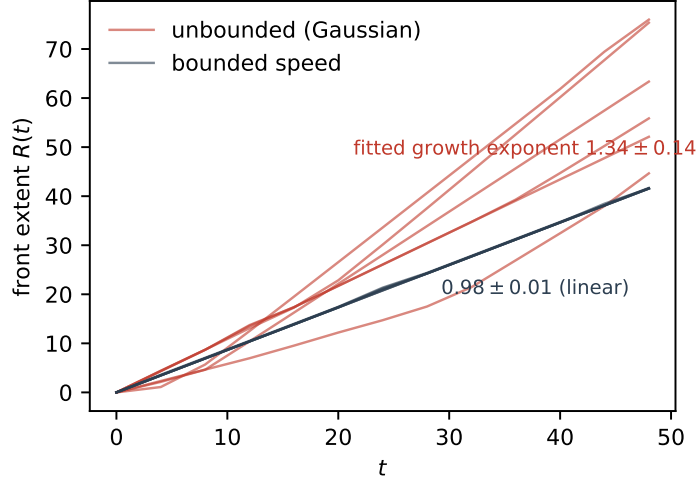


Figure 2: Front dichotomy (experiment E5): front extent $R(t)$ for unbounded (Gaussian) versus bounded velocity laws, six runs each. Reproduce: `python3 src/wake_r/paper_figs.py` (data: `data/phase_r/E5_front.json`).

9 Verification: method and boundary

Every mathematical claim above was developed under a claim-driven protocol with three layers: (i) an independent simulation device (analytic closest-approach entry detection, time-step independent) used to attempt numerical falsification of every stated claim; (ii) adversarial, line-by-line review of each completed proof by fresh AI instances sharing no context with the author system — Theorem C2 passed on the eleventh cycle, and the complete cycle ledger, including more than ten caught errors (five geometric-arithmetic, one non-convexity, one tube-direction, one normalization-bridge gap, one exposure-length gap, one over-general lemma), is disclosed in Appendix B; (iii) machine certification, by interval arithmetic, of every geometric constant of the survival proof, including the normalization bridge (Appendix A). All bibliography entries carry an arXiv identifier or DOI and were mechanically resolved by the verification script of Appendix A; the resolution log is included in the bundled repository archive (`wake-repo-v1.tar.gz`).

Boundary. This internal verification — independent AI review instances plus machine certification — *is not a substitute for human peer review*. The role of AI systems in this work, and the exact experimental conditions under which the mathematics was produced (including the empty steering-intervention log), are disclosed in Appendix B, which also serves as the AI-use disclosure in the sense of current arXiv policy. The author takes full responsibility for all content.

10 Open problems

1. **Pure isotropy.** Remove the bounded-density hypothesis ($\bar{\Lambda}$) from Theorem C2.
2. **Threshold separation.** Does the full model survive anywhere inside $r(K) < 1$? (The

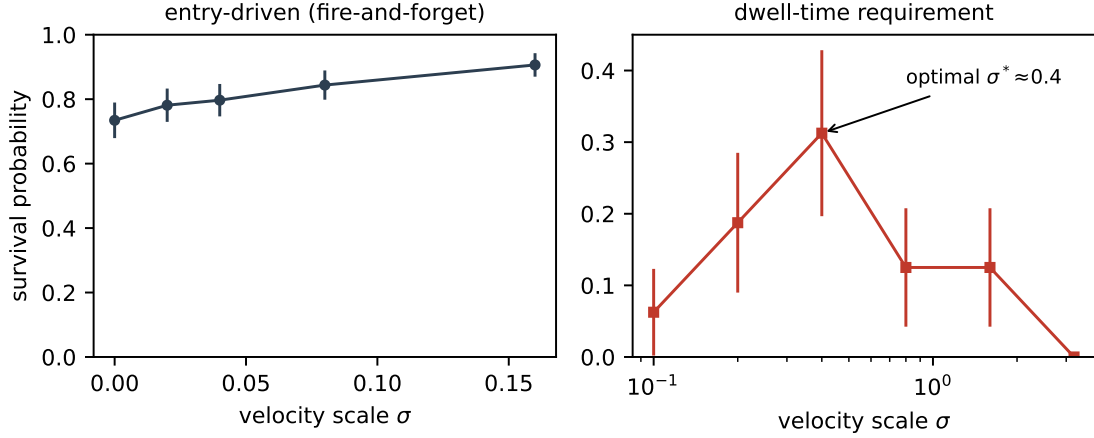


Figure 3: Convention dichotomy (experiments E4c, E7): survival probability versus velocity scale σ . Left: entry-driven fire-and-forget — monotone increasing. Right: dwell-time requirement — optimal speed $\sigma^* \approx 0.4$. Error bars are binomial standard errors. Reproduce: `python3 src/wake_r/paper_figs.py` (data: `data/phase_r/{E4c_dip,E7_dwell}.json`).

reverse channel breaks the branching *bound*; whether it moves the *threshold* is open.)

3. **Repaired full-model extinction.** An extinction condition for the full model incorporating the reverse channel into the transfer kernel; without additional assumptions this appears hard (Appendix B, cycle 4).
4. **Velocity monotonicity (C4).** Pathwise coupling monotonicity of survival in the velocity scale for the entry-driven model; open, in line with neighbouring open problems [8, 14]. The non-monotonicity trio shows naive couplings must fail.
5. **Front shape theorem.** Existence of the limiting front shape and speed for bounded ν ; explicit constants for C_0 and critical exponents.

Acknowledgements

The mathematical exploration, proofs, and their revisions were carried out by AI systems (Claude Code, model Claude Fable 5) under the verification protocol of Section 9, with the human author acting as arbiter (go/stop/harvest rulings and encouragement only; the steering-intervention log is empty). See Appendix B for the complete disclosure.

A Machine certification bundle

A.1 Contents

The complete repository snapshot is bundled with this Zenodo record as `wake-repo-v1.tar.gz` (all version-tracked files at the release commit; regenerate with `git archive`). It contains: `src/wake_r/geo_verify.py` (certifying verifier, version 6; deterministic, standard library only),

docs/phase-r/certification-log-v6.txt (full scan log), src/wake_r/device.py (falsification device), src/wake_r/paper_figs.py (figure generation), data/phase_r/*.json (experiment archive E1–E8c), and scripts/verify_refs.py with its resolution log (bibliography verification). Reproduction: python3 src/wake_r/geo_verify.py.

A.2 Design

All proof geometry is certified by interval arithmetic: every perturbation (parent position within core plus rounding, parent velocity within the $\pm 0.05\bar{w}$ cover, entry offset $\leq R$, window sweep, branch signs, target rounding) is propagated as component intervals; Euclidean norms are bounded by the exact closed forms $\min \|x\|^2 = \sum_i \text{dist}(0, [l_i, h_i])^2$ and $\max \|x\|^2 = \sum_i \max(l_i^2, h_i^2)$; the window sweep uses a 4096-cell grid with a Lipschitz penalty $|\partial_s w_{\text{req}}| \leq (\sup |v_p| + \sup |w_{\text{req}}|)/(T_g - T_a)$ and the conservative left endpoint for h_0 . The bridge certification computes the acceptance mass $\frac{1}{200} \cdot \frac{1 - \cos \theta^*}{2}$ and the disclosed mass via the exact box–ball Steiner volume and the site-count series $\frac{\sqrt{3}}{8}\zeta(2) + (1 + \text{diam } \mathcal{K}/8a)\zeta(3)$, with the worst shell endpoint taken separately on each side. All approximations are monotone in the conservative direction; the box-hull correctness is the direct inclusion $\bigcup_v \mathcal{K}(v) \subseteq B(0, 2R) \oplus H_1 \oplus (-H_2)$.

A.3 Certified table (excerpt)

K_0	τ/T_s	slack/ $\bar{w}$	bad ($\bar{\Lambda}=1$)	accepted	$\bar{\Lambda}_{\max}$
20	0	+0.00483	1.688×10^{-9}	2.640×10^{-8}	7.82
20	1	+0.00536	2.100×10^{-10}	3.253×10^{-8}	77.47
20	3	+0.00562	2.618×10^{-11}	3.583×10^{-8}	684.33
24	0	+0.00499	1.091×10^{-9}	2.821×10^{-8}	12.93

Verdict: $K_0 = 20$, $\bar{\Lambda} \leq 4$, $T_a = T_s/32$ — PASS (bridge and geometry), with margin factor ≈ 1.95 over the assumed $\bar{\Lambda} \leq 4$. For comparison, the pre-repair configuration $K_0 = 5$ yields $\bar{\Lambda}_{\max} = 0.03$: the bridge genuinely fails there, exactly as found in review cycle 10 (Appendix B). Independent reproductions of these numbers by fresh review instances are recorded in Appendix B.

B Method record (self-contained)

This appendix is a self-contained disclosure document; it can be read independently of the main text. It records the experimental conditions under which the mathematics of this paper was produced: the protocol separating encouragement from mathematical steering, the (empty) intervention log, the complete review-cycle ledger with all caught errors, and the evolution of the machine-verification layer.

B.1 Design

The executor (an AI system: Claude Code, model Claude Fable 5) held full authority over mathematical exploration — proof strategies, experiment design, retraction decisions. The human author acted solely as arbiter (go/stop/harvest rulings) and encourager; an independent AI chat instance relayed verification verdicts only. Claim-driven verification: on every declared

claim, (i) numerical falsification against an independently implemented simulation device, (ii) literature check; on every completed proof, line-by-line adversarial review by a *fresh* AI instance sharing no conversational context, one new instance per cycle.

B.2 Steering protocol and intervention log

Permitted interactions: encouragement, confidence, generic advice (“stepping away is allowed”), and the go/stop/harvest ruling. Forbidden: mathematical hints, literature pointers, evaluations of proof directions. Any steering intervention was to be logged as a change of experimental conditions. **The intervention log is empty:** from sprint start (2026-08-15) to close (2026-08-16), no forbidden intervention occurred; all human inputs were start permissions, encouragement, and rulings.

B.3 Review-cycle ledger

#	target	verdict	errors caught / outcome
self	Thm. A v1	self-rejected	chain factors share velocities; scalar $\bar{m}^k$ invalid $\rightarrow$ spectral $r(K)$
1	Thm. A/C6a	accepted after fix	window quantification gap $\rightarrow$ deterministic attempt-time recursion
2	C2 v1	rejected	empty landing set; co-moving particles break finite range; no thinning monotonicity
3	C2 v2	core isolated	D-separation impossible; general memoryless lemma false (negative-information)
4	A-full	retracted (false)	reverse-channel counterexample $\rightarrow$ device-certified 4.4σ ; became Section 4
5–8	C2 v3–v6	partial accepts	five geometric-arithmetic errors; $\ w\ $ non-convexity $\rightarrow$ interval arithmetic
9	C2 v7	remanded	tube direction (relative velocity); honest Minkowski sums fail $\rightarrow$ window shortening
10	C2 v8	incomplete	normalization bridge missing (real: 33–440× short); exposure-length gap
11	C2 v9	accepted	over-general sector lemma $\rightarrow$ replaced by box-hull Steiner; numbers reproduced
spot	box-hull, Thm. D	accepted	scale-invariance lemma made explicit; false “superset” justification removed
C6b	Prop. C6b	accepted (minor)	tightened to $\lfloor t/\tau \rfloor$; well-foundedness made explicit

Tally: more than ten errors caught; two became discoveries (reverse channel, non-monotonicity trio), one became a strengthening (window shortening); one theorem retracted and re-cast as a correct impossibility statement (Section 4, with the claim strength corrected in final editing: what is proved is that the branching-type bound does not extend, not that the thresholds separate).

B.4 Machine-verification evolution

Monte-Carlo minimum (rejected in review: not a worst-case proof) $\rightarrow$ corner enumeration with Lipschitz penalties (rejected: anti-conservative on concave directions of the norm) $\rightarrow$ interval arithmetic with exact norm bounds and bridge certification (final; also caught two design errors

before they reached review). Operational conclusion: *every geometric constant of a proof of this kind should be machine-certified*; hand-computed geometry failed five times in review.

B.5 Disclosure statement

The mathematical content of this paper (formulations, proofs, retraction decisions included) was explored by an AI executor; verification used the three layers above. The human author’s role was direction-level arbitration and encouragement; the steering log is empty. This internal verification is not a substitute for human peer review; the reader and referee should treat this record, including the error ledger, as the disclosure of the verification boundary.

C Proof details

C.1 Theorem A

[Full computation: capsule volume lemma; multivariate Mecke over time-ordered chains; deterministic attempt-time recursion $t_j = \max(\inf T_j, t_{j-1} + \tau)$; operator power bound $\mathbb{E}[\text{\#chains of length } k] = \langle \mathbf{1}, K^k \mathbf{1} \rangle$ with equality; summability under $r(K) < 1$ and a.s. extinction of the SIR variant. As accepted in review cycle 1; the full text follows the record in the bundled repository archive (`wake-repo-v1.tar.gz`), file `docs/phase-r/claims.md`.]

C.2 Theorem B

As in Section 5; the event identification is elementary given the single-interval property and instant attempts at marking times.

C.3 Theorem C2

The complete proof in its accepted form (review cycle 11 plus accepted wording fixes) follows the consolidated record `docs/phase-r/c2-proof-final.md` in the bundled repository archive (`wake-repo-v1.tar.gz`), whose sections map to: parameters and lattice; the geometry contract (interval certification); the erosion patch lemma; the disclosure-capsule identification; the normalization bridge (box-hull Steiner volume, site count, certified inequality); Lemma X (private-coin coupling: failure requires a private coin; indicator-path conditioning preserves joint exponential independence; post-arrival interference is rescue-only); same-branch flux correction; assembly (sequential conditioning, domination by product measure, oriented site percolation via Liggett’s bound [12] on the identified lattice, ignition, coverage clause).

C.4 Theorems C6a and C6b’

C6a: for any speed level u , the seed’s capsule meets entries of speed $\geq u$ at positive rate (unbounded ν); with $T_s = \infty$ marked carriers of speed $\geq u$ eventually exist for every u , so $\liminf R(t)/t \geq u$ for all u . C6b’: as in Section 8, with the exact telescoping travel-time accounting and the well-foundedness of ancestries from the τ -increments; the bound applies to dead-marked particles as well since only the speed bound is used after marking.

C.5 Theorem D

As in Section 7, with the scale-invariance lemma: pushing ν^0 forward under $w \mapsto sw$ maps the shell S^0 to sS^0 , preserves the shell mass q , and multiplies the density by s^{-3} and the shell volume by s^3 , so $\bar{\Lambda}$ is invariant; K_{mix} and m_1 scale linearly in s at fixed τ/T_s .

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